Łukasz Tychoniec, Ph.D.

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Research Profile

Observational astrophysicist with over 10 years of research experience. Expert in using the Atacama Large Millimeter/submillimeter Array (ALMA) and James Webb Space Telescope (JWST) to observe the protostellar systems at the onset of planet formation. Published 85 peer-reviewed works (7 as the first author) with over 3000 citations (h-index: 32); invited to over 10 seminars and talks, and supervised over 10 MSc, BSc, and internship students. My research combines studies of early star formation with planet formation and evolution, placing me at the forefront of characterizing young, forming stars and their planet-forming disks with state-of-the-art observatories. Recipient of the 2025 Young Astronomer Award of the Polish Astronomical Society.

Leadership experience

Y-Disks collaboration - principal investigator

Young Disks (Y-Disks) is a JWST Cycle 4 program on observing 16 protostars with the MIRI instrument, awarded 24.5 hrs of telescope time. As the project PI, I led proposal preparation and currently coordinate a team of over 20 researchers. The observations were delivered in February 2026, and the first publications from the project will be submitted by the end of 2026.

JOYS collaboration - sub-group leader

JWST Observations of Young protoStars (JOYS; PI: Ewine van Dishoeck) is a collaboration of more than 30 researchers, working on JWST Guaranteed Time Observations of the MIRI European Consortium. I led the data acquisition and preparation; currently, I lead data reduction efforts and the outflows and jets subgroup. The collaboration published more than 20 publications, including 2 with me as the first author.

I-HOW IAU JWST Hands-On Workshop 2025 - scientific coordinator

I coordinated the organization, funding acquisition, and preparation of the workshop at NCU in Toruń in July 2025 on JWST data analysis for graduate students. I coordinated a team of tutors to run a two-week workshop with 33 participants from 8 countries, with a budget of 75 000 €.

Lorentz Center Workshop - main coordinator

I led an effort to acquire funding and organize an international workshop on star and planet formation using JWST. The 5-day workshop with 50 participants was held in Leiden at the Lorentz Center, where I served as the chair of the Scientific Organizing Committee.

Education

Ph.D. in Astronomy; Leiden University, The Netherlands
Sept 2016 – Oct 2020

M.Sc. in Astronomy; Adam Mickiewicz University, Poznań, Poland
Oct 2014 – July 2016

B.Sc. in Astronomy; Adam Mickiewicz University, Poznań, Poland
Oct 2011 – June 2014

Awards & Scholarships

Awards

- **Young Astronomer Award** of the Polish Astronomical Society 2025
- **ESO Fellowship**, Garching, Germany 2020
- **LEAPS Summer Studentship**, Leiden, The Netherlands 2015
- **Scholarship**, Polish Minister of Science and Higher Education 2015
- **Scholarship**, Rector of Adam Mickiewicz University, Poland 2015

Funding

- **Lorentz Center** grant for workshop organization 21 000 EUR
- **NOVA** grant for workshop organization 4 000 EUR
- **IAU** Funding for the I-HOW workshop 35 000 EUR
- **Polish Ministry of Science and Higher Education** funding for the I-HOW Workshop 16 000 EUR
- **PROM** funding for the I-HOW Workshop 25 000 EUR
- **FINCA fellowship** (declined) 100 000 EUR
- **PSF fellowship at MPIA Heidelberg** (declined) 140 000 EUR

Telescope time as a Principal Investigator

- **Atacama Large Milimeter/Submillimeter Array (ALMA)** 136 hrs total
- **James Webb Space Telescope (JWST)** 24.5 hrs total

Professional Services (selected)

- **Reviewer:** FONDECYT Chilean National Grant 2026
- **Scientific Coordinator:** I-HOW JWST Hands-On Workshop for Central & Eastern Europe July 2025
- **Chair of the Scientific Organizing Committee** Lorentz Workshop: Protostellar accretion and ejection with JWST and beyond May 2025
- **Guest Editor** Frontiers in Astronomy and Space Science 2024-2025
- **External Reviewer** JWST Cycle 4 TAC 2024
- **Scientific Organizing Comittee member** European Astronomical Society Session June 2024
- **Panel Member:** NASA ROSES 2023 Evaluation Panel 2023
- **SOC Member:** 6th Shaw Workshop 2024
- **Jury:** I edycja Poznań Science Slam 2024
- **Scientific Assistant:** Observing Program Committee at ESO 2021–2022
- **Mentor:** DUSTBUSTERS School May 2022
- **Peer Review:** Referee for MNRAS, ApJ, and A&A since 2019

Selected Presentations

Invited Talks

- *European Astronomical Society meeting*, Padova, Italy (June 2024): "Protostellar disk masses: estimating reservoir of planet-forming material"

- *Raising the veil on star formation near and far*, Cambridge, UK (April 2024) : "Astrochemical diagnostics of protostellar sources: from ALMA to JWST"
- *Star@Lyon 2023*, Lyon, France (June 2023): "Protostellar disc masses: estimating reservoir of planet-forming material"
- *VLTI/ALMA Interferometry Workshop*, Garching, Germany (Jan 2023): "RESOLVE algorithm applied to ALMA observations"
- *MIAPP Workshop*, Garching, Germany (Oct 2021): "Chemical map of the Solar System analogues in the embedded stage"